

A reproducible photonic pilot delivers a useful negative result

Few-label steel defect classification · ideal photonic simulation
One outer-fold pilot · frozen protocol · measured on 11 Sep 2026

PHOTONIC Q1

0.8708

macro-F1 · 48 labels per class

FULL CLIP + LINEAR

0.9852

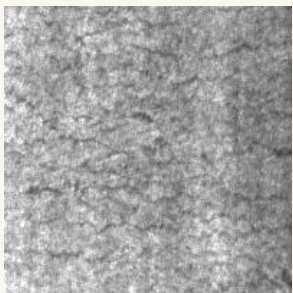
macro-F1 · same test fold

Q1 – B1

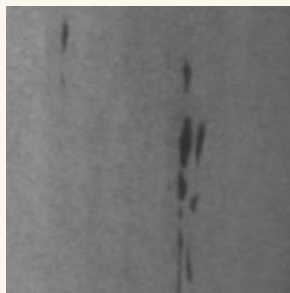
-0.1143

Q1 beats locked RFF — but not ELM, RBF-SVM, or the full embedding.

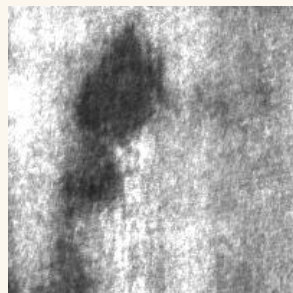
The benchmark starts with 1,800 audited images



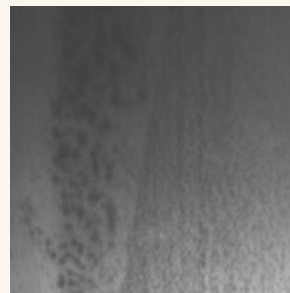
Crazing



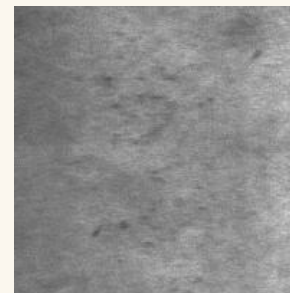
Inclusion



Patches



Pitted surface



Rolled-in scale



Scratches

AUDIT RESULT

1,800 / 1,800

valid images · 0 quarantined

6 × 300

balanced classes

ONE EXACT DUPLICATE PAIR

Pa_101 · Pa_105



SAME GROUP
never split

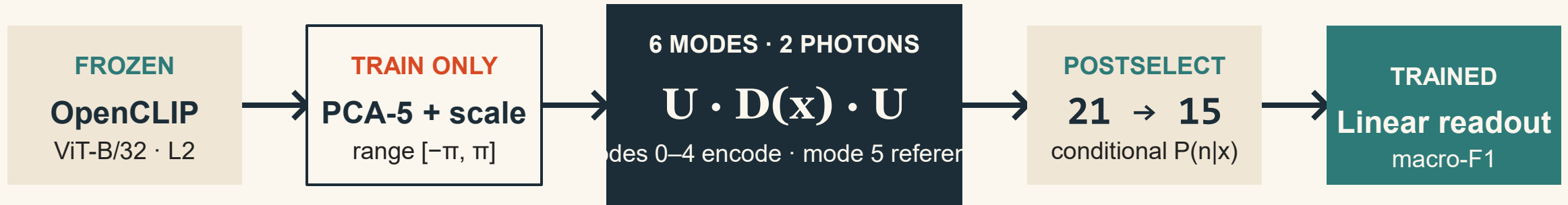


GROUP-SAFE FOLD 0
60 test / class
seed 42

No sample-ID or duplicate-group overlap
across train · validation · test

Source: dataset_audit.csv · frozen experiment protocol

The circuit is fixed; only phase encoding changes



FIXED PER MAP SEED

Same Haar-random U on both sides

Seeds 101 · 202 · 303 are averaged

They are never selected or ensembled.

Input occupation

$[1, 0, 1, 0, 0, 0]$

$$U(x) = UD(x)U, \quad \phi_Q(x) = \text{Normalize}(\{P(\mathbf{n} | x)\}_{\mathbf{n} \in \mathcal{C}})$$

INDEPENDENT IMPLEMENTATION CHECKS

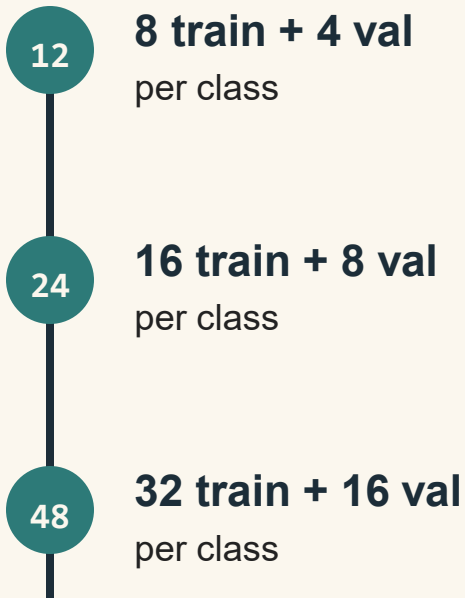
Perceval 1.2.4 · 4.16×10^{-17} max error

MerLin 0.4.1 · 8.33×10^{-17} max error

Source: experiment_protocol.md · verification.json

The protocol makes overclaiming difficult

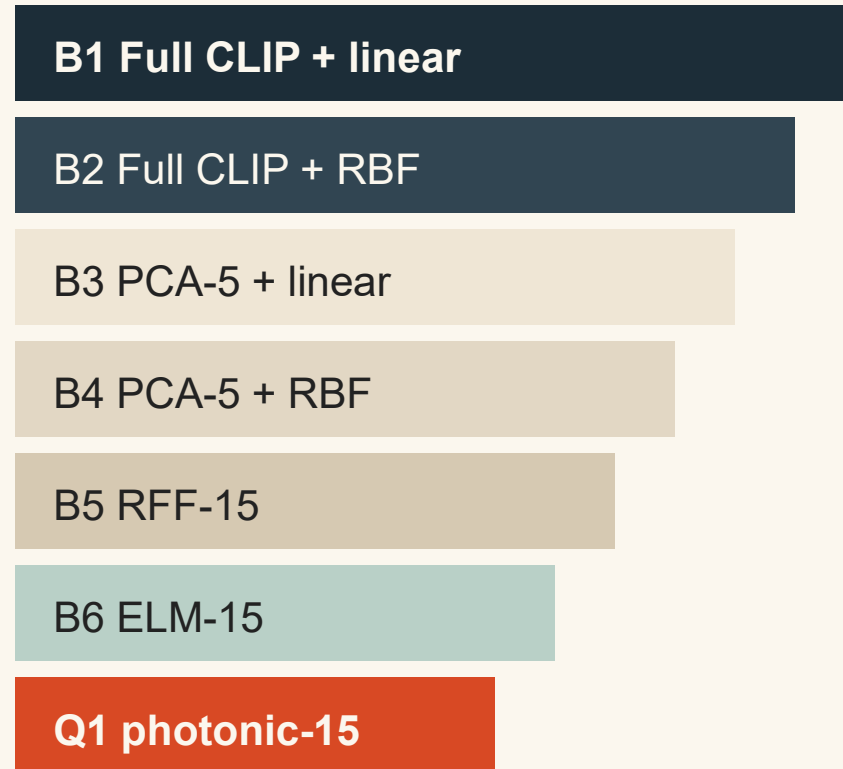
PAIRED LABEL BUDGETS



SUBSET SEEDS 11 · 22 · 33

Nested, same outer test fold

SEVEN FAMILIES · SAME TEST GATE



R1 LEDGER

1,404
candidate fits

117
test evaluations

SELECTION RULES

Validation macro-F1 only

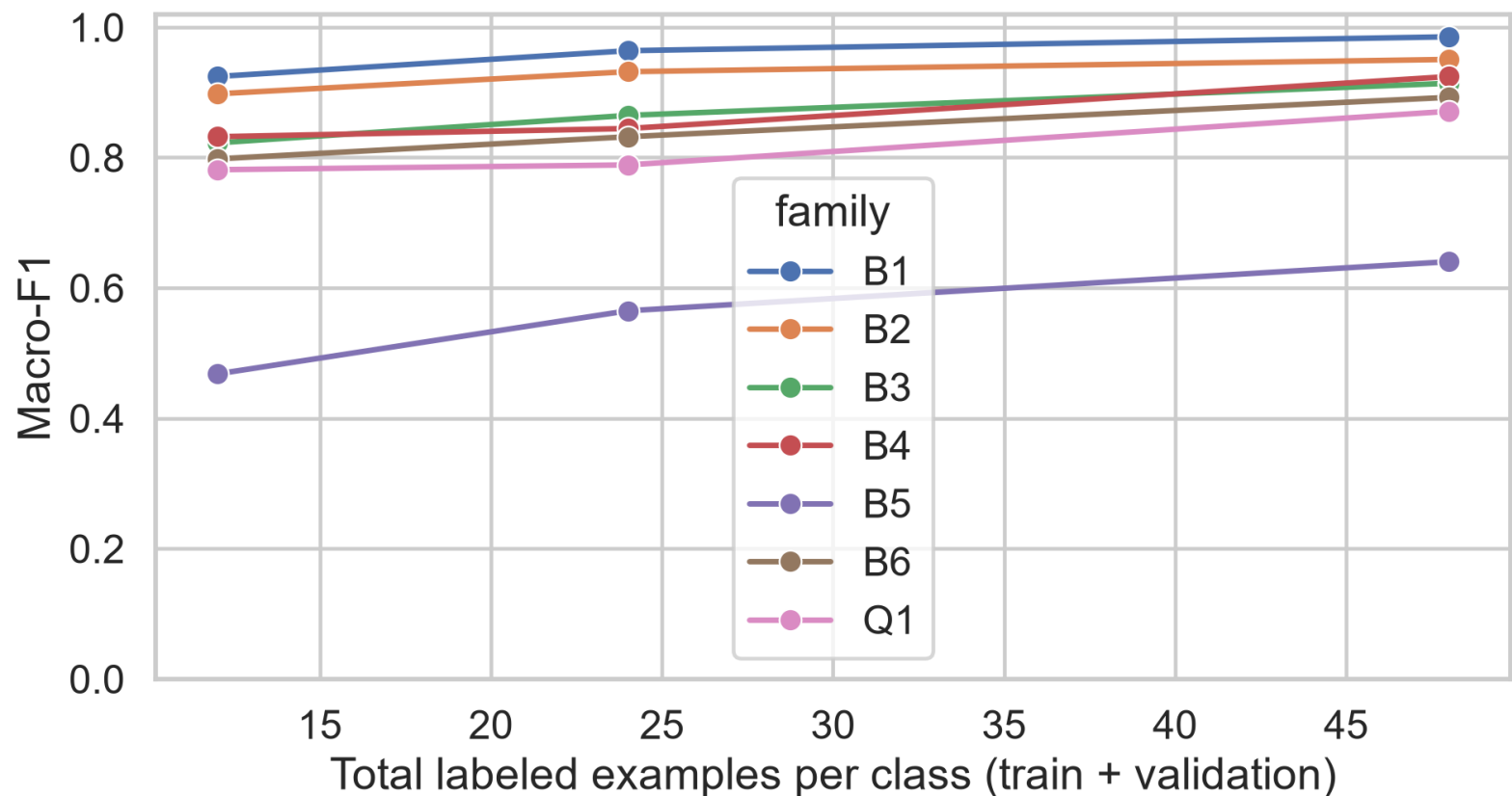
Map seeds averaged—not picked

No train + validation refit

One locked test evaluation

Source: frozen experiment protocol · repeated evaluations share outer test data

Q1 improves with labels, but full embeddings dominate



48 LABELS / CLASS

0.8708

Q1 macro-F1

0.9852

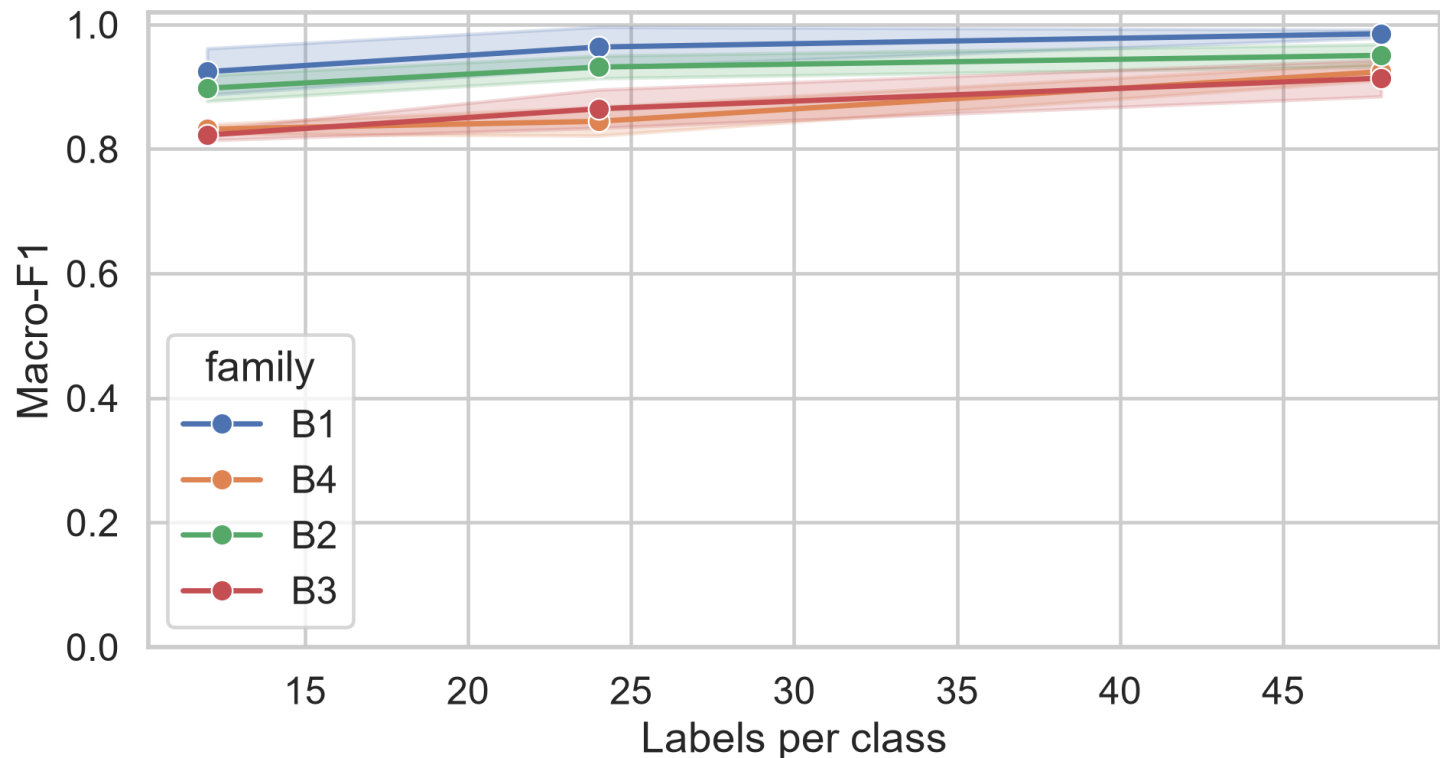
B1 · full CLIP + LR

WHAT SURVIVES

Useful structure is learned.

No case for discarding
the full embedding.

PCA-5 compression explains much of the gap



FULL → PCA-5 PENALTY

Linear: B1 – B3

+.1018 +.0993 +.0713

12 24 48 labels

RBF: B2 – B4

+.0657 +.0872 +.0263

AT 24 LABELS / CLASS

B3 0.8646

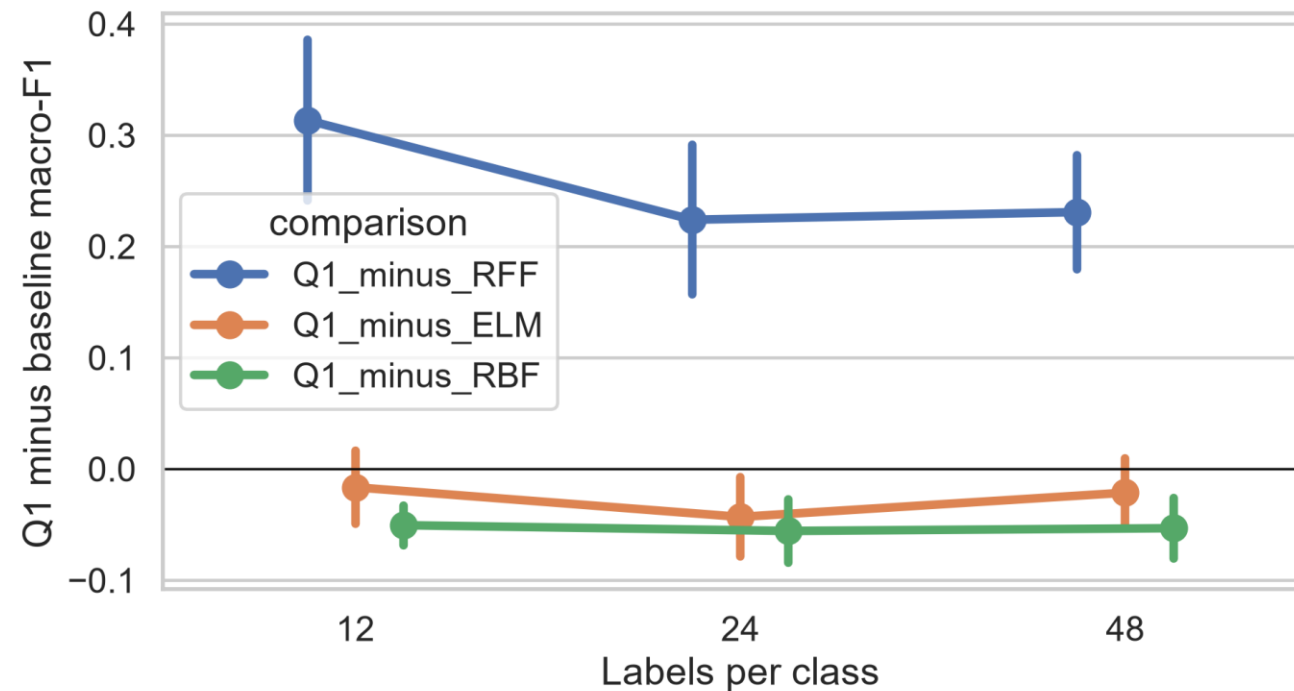
B4 0.8444

B6 0.8318

Q1 0.7886

Signal is lost before the
nonlinear map is evaluated.

Q1 beats RFF—not the strong nonlinear controls



Q1 – RFF

+.3133 · +.2238 · +.2307

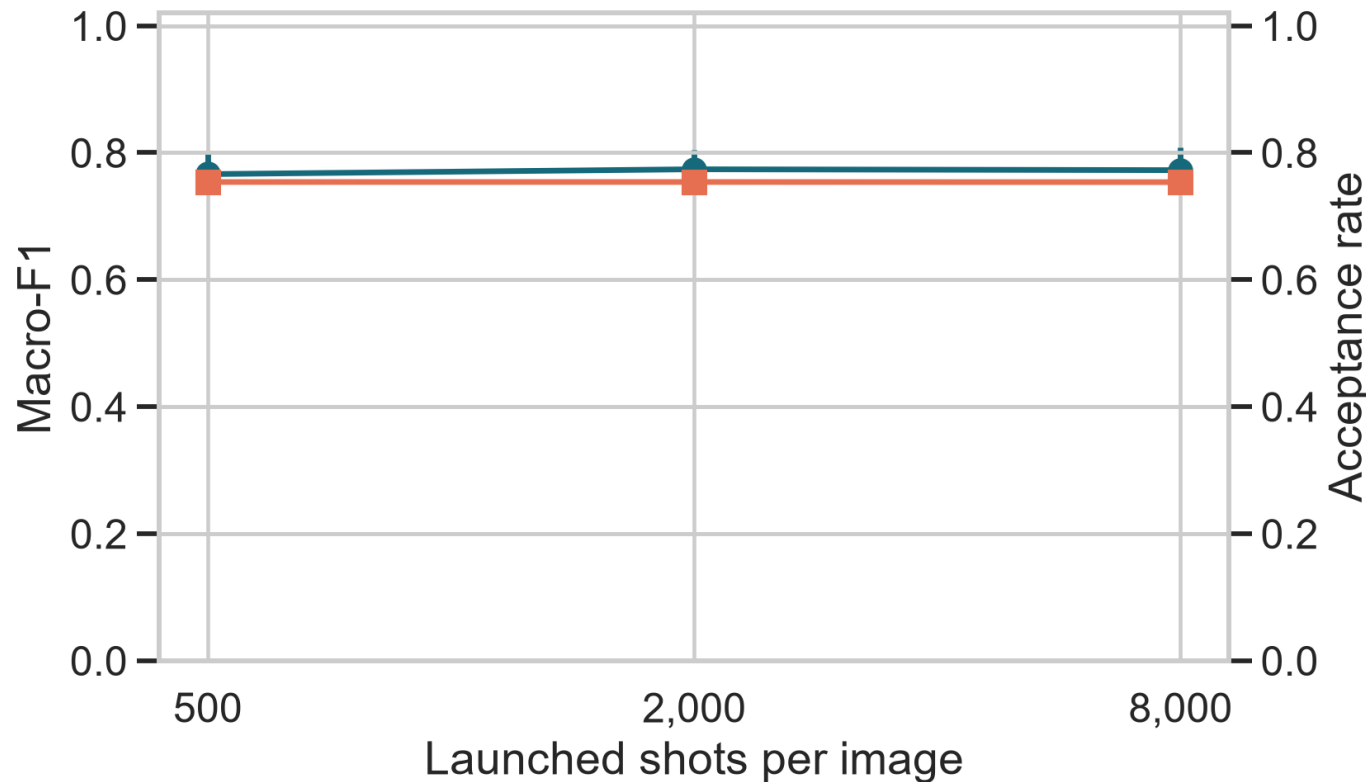
Q1 – ELM

-.0166 · -.0432 · -.0215

Q1 – PCA-5 RBF

-.0507 · -.0558 · -.0534

More launched shots do not resolve the model gap



MACRO-F1 · 15 EVALUATIONS EACH

500 0.7662 ± .0322

2,000 0.7739 ± .0308

8,000 0.7725 ± .0363

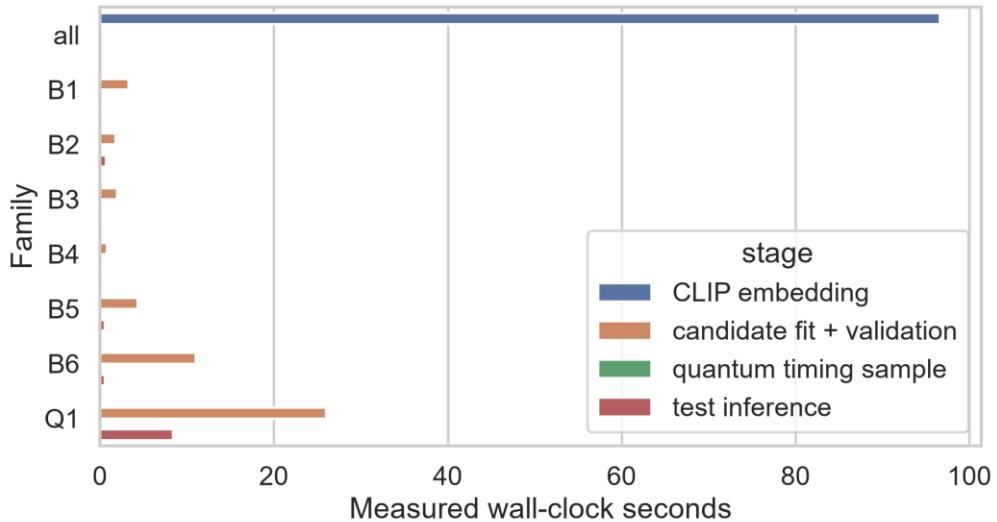
acceptance ≈ 0.754

0 zero-acceptance failures

CAVEAT

Ideal-trained readout;
not noise-aware training.

Scientific validity—not runtime—is the bottleneck



MEASURED ON CPU

OpenCLIP extraction **96.57 s**
R1 after embeddings **≈59.9 s**

Local CPU profiling only · no loss, phase drift, detector noise, or QPU queue

EVIDENCE LADDER

1 · Simulator semantics validated

2 · Repeated subset evaluations

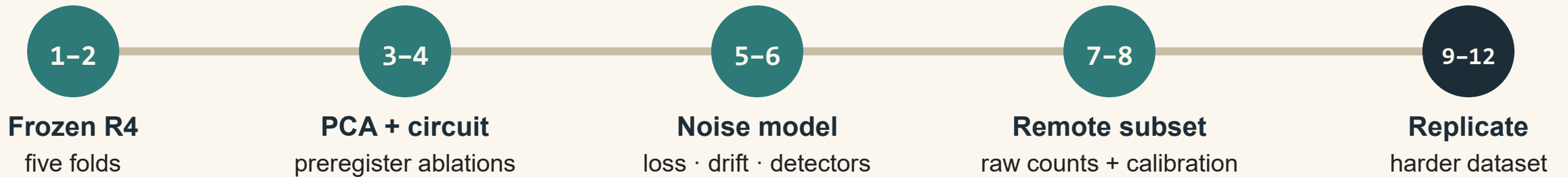
3 · Outer-fold generalization

4 · Realistic photonic noise

5 · Hardware execution

One outer fold · small validation cells · B1 near ceiling

Continue only if a falsifiable gate can change the decision



CONTINUE ONLY IF

Q1 closes the ELM / RBF gap across outer folds

or demonstrates a resource tradeoff with practical value.

STOP CONDITION

If Q1 only beats RFF:
feasibility—without utility.

Repository · github.com/MonsieurNam/Quanova_Photonic_Defect_QML